

Project Initialization and Planning Phase

Date	6 th July 2026
Team ID	SWTID-2026-4078
Project Title	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This project automates the credit card approval process using machine learning. It analyses applicant financial and demographic information to predict whether a credit card application should be approved or rejected. The solution reduces manual effort, improves consistency, speeds up decision-making, and minimizes the risk of human error.

Project Overview	
Objective	Develop an intelligent machine learning system that predicts whether a credit card application should be approved or rejected based on applicant information with high accuracy.
Scope	The project includes data collection, preprocessing, feature engineering, model training, evaluation, and deployment using Flask. Multiple machine learning algorithms are compared to identify the best-performing model for real-time prediction.
Problem Statement	
Description	Banks receive thousands of credit card applications every day. Manual verification is time-consuming, inconsistent, and prone to human error, resulting in delayed approvals and increased operational costs.
Impact	Automating the approval process improves decision-making speed, increases prediction accuracy, reduces financial risks, minimizes manual effort, and enhances customer satisfaction.
Proposed Solution	
Approach	Historical applicant data is pre-processed and used to train Machine Learning models including Logistic Regression, Decision Tree, Random Forest, and XGBoost. The best-performing model is integrated into a Flask web application for real-time prediction.

Key Features	• Automated Credit Card Approval Prediction • High Accuracy using Machine Learning • Real-Time Prediction through Web Application • User-Friendly Interface • Fast Decision-Making Process • IBM Watson Machine Learning Deployment Support • Secure and Scalable Architecture
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Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor (4+ Cores)
Memory	RAM specifications	8 GB RAM (Minimum), 16 GB Recommended
Storage	Disk space for datasets, trained models, and application files	512 GB SSD / 1 TB HDD
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost
Development Environment	IDE	Jupyter Notebook, Visual Studio Code, PyCharm
Data		
Data	Source, Size, Format	Credit Card Approval Dataset (Kaggle), CSV Format, ~20,000+ Applicant Records